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SymphonySynth: Melody Generation using a Hybrid Approach

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ABSTRACT

Music generation through artificial intelligence has evolved beyond conventional deep learning models into hybrid intelligent systems that blend symbolic reasoning with data-driven sequence modeling. SymphonySynth is a hybrid melody generation framework integrating rule-based pre-processing, a Transformer-based symbolic melody learner, and Mamba sequence modeling for long-term coherence. The system tokenizes MIDI compositions using the REMI-M (Rhythm, Event, Measure, Instrument – Modified) format to capture temporal, harmonic, and structural dependencies. The hybrid architecture combines Transformer's strong pattern learning with Mamba's state-space sequence modeling, supported by music theory rules to maintain key, scale, and rhythm consistency. Preliminary experiments with general MIDI datasets demonstrate coherent short melodies. The model will be fine-tuned on Indian Raga datasets to enable culturally adaptive melody synthesis.

INTRODUCTION

Music generation has evolved significantly with advancements in deep learning, enabling systems to compose realistic and musically coherent melodies. However, generating traditional and culturally rich melodies such as Indian ragas or Chinese folk tunes presents a unique challenge due to their intricate ornamentations, tonal sensitivity, and rhythmic complexity.

Our project, titled "SymphonySynth: Melody Generation using a Hybrid Approach," focuses on creating a hybrid architecture combining Mamba State-Space Models (SSMs) and Transformers to generate melodies that respect traditional scales and phrasing. Using MIDI-based tokenization (REMI-M), the system converts musical patterns into structured symbolic sequences, allowing the hybrid model to capture both short-term rhythmic patterns and long-term melodic dependencies effectively.

STATEMENT OF PROBLEM

Traditional melody generation models often fail to reproduce the expressive nuances of cultural music, especially ragas, which rely on microtonal variations, ornamentations (gamakas), and specific ascending/ descending rules. Furthermore, datasets for such traditional music are limited, leading to poor generalization in deep learning models. The problem is to develop a system capable of understanding and generating traditional melodies with cultural and structural accuracy, even from limited data, while maintaining musical coherence, tonal consistency, and rhythm fidelity.

OBJECTIVES

1. Design a hybrid deep learning framework combining Transformers and Mamba for expressive melody generation.
2. Develop a tokenization scheme (REMI-M) for structured event representation.
3. Integrate rule-based filters for tonal, rhythmic, and tempo consistency.
4. Train on general MIDI datasets and fine-tune on Indian Raga compositions.
5. Evaluate melodies using statistical metrics and subjective listening tests.

SYSTEM DESIGN AND ARCHITECTURE

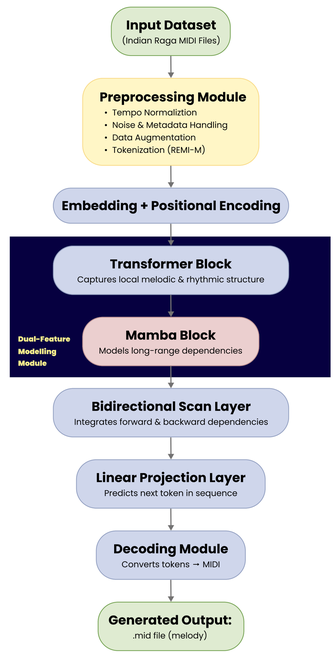

Fig 1: System Architecture

PROPOSED METHODOLOGY

The workflow of SymphonySynth consists of four main stages:

1. Data Preparation
   - Tempo Normalization: Ensures consistent rhythmic structure across MIDI tracks.
   - Noise and Metadata Handling: Removes non-musical artifacts and processes instrument or key information.
   - Data Augmentation: Expands the dataset through pitch shifting and time stretching.
   - Tokenization with REMI-M: Converts MIDI files into symbolic tokens while incorporating Indian gamakas and ornamentations.
2. Model Training
   - Sequential Hybrid Design: The Transformer learns short-term temporal features; its output is sequentially fed into Mamba for long-range context.
   - Regularization & Optimization: Techniques such as dropout and adaptive learning are used to stabilize training.
   - Handling Challenges: Addresses limited dataset size, tonal sensitivity, and rhythmic complexity inherent in traditional music.
3. Fine-Tuning and Evaluation
   - Fine-Tuning Dataset: The pre-trained hybrid model is fine-tuned on the Music.Traditional.HinduRaga.MIDI dataset.
   - Sequence Generation: The trained model generates token sequences, which are decoded back into MIDI.
   - Evaluation Metrics:
     - Objective: Cross-entropy loss, token diversity, tonal stability.
     - Subjective: Listener feedback on melodic coherence and cultural authenticity.
4. Output
   - Produces a generated melody in MIDI format, playable through any digital audio workstation (DAW) or synthesizer.
   - Represents the final synthesis stage of the SymphonySynth workflow.

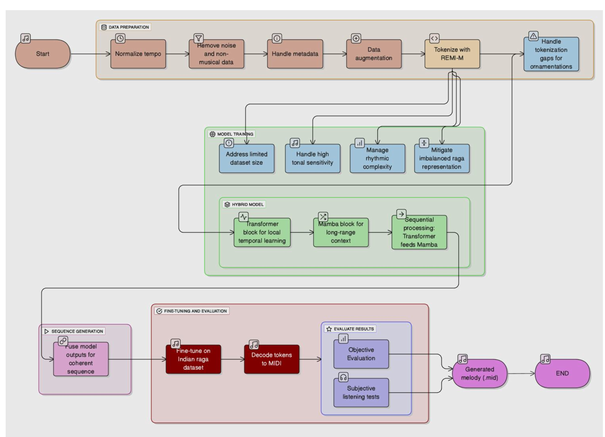

Fig 2: Workflow of SymphonySynth

RESULTS

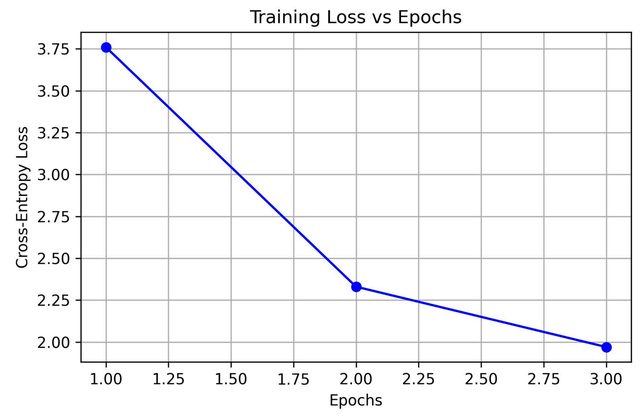

Fig 3: Loss Curve

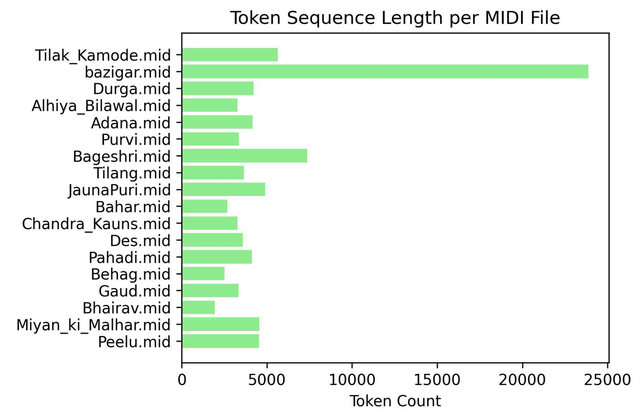

Fig 4: Token Distribution

CONCLUSION

The SymphonySynth project successfully establishes a hybrid music generation framework that bridges deep learning and symbolic reasoning. With the integration of Transformer and Mamba models, the system can handle both short-term phrasing and long-term context efficiently. The addition of rule-based post-processing ensures generated melodies remain musically valid within a desired raga or scale. Future work includes expanding the dataset, introducing human-feedback-based reinforcement, and real-time melody synthesis through interactive interfaces.

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